

Community Security Alert System (CSAS): Systems Analysis and Design — Workshop No. 1

Felipe Jose Garzon Herrera — 20251020132
Quintero Gordillo Juan Esteban — 20251020137
Henry Samuel Garrido Medina — 20251020125
Gabriel Mateo Cusba Marin — 20251020128
Computer Engineering Program
Universidad Distrital Francisco José de Caldas
Bogotá D.C., Colombia — Semester 2026-I

Abstract

This report presents a complete systems analysis of the Community Security Alert System (CSAS), a crowd-sourced platform designed to improve safety on university campuses. The document covers system scope and boundary definition, stakeholder identification, operational environment characterization, data collection planning and execution, data quality evaluation, system element decomposition, relationship analysis, and a sensitivity study of key behavioral parameters. Primary data was collected through structured interviews with campus security staff, student surveys ($n = 20$), direct observation across four campus zones (8 sessions total), document analysis, and benchmarking of existing security platforms in Bogotá. Findings consistently show that night-time incident concentration, a low community reporting culture, and the absence of a centralized mobile reporting tool are the main gaps the CSAS is designed to address.

Index Terms

campus security, community alert system, systems analysis, crowd-sourced incident reporting, real-time notifications, data collection, sensitivity analysis, feedback loops

I. SYSTEM OVERVIEW AND SCOPE DEFINITION

A. System Description

The Community Security Alert System (CSAS) is a collaborative technological platform designed to enhance safety within university campuses and their immediate surroundings. By integrating real-time notifications with crowd-sourced incident reporting, the system empowers students, faculty, and staff to act as active sensors for community safety. Users can report incidents—ranging from medical emergencies and accidents to suspicious activities—via a mobile or web interface. To maintain data integrity, the system employs verification mechanisms including multi-user confirmation and administrative moderation, ensuring that alerts are both high-speed and high-reliability.

B. System Boundaries

Defining clear boundaries is essential for scoping the analysis and preventing scope overextension in subsequent design phases. Table I separates what falls within the CSAS from what constitutes its external environment.

TABLE I: *System Boundary Definition — CSAS*

Inside the system (scope)	Outside the system (environment)
Incident reporting & management platform	University physical infrastructure
Crowd-sourced verification logic	Local law enforcement (Police / EMS)
Real-time push notification engine	Public telecommunications & ISP networks
User and incident databases	Non-registered external visitors
Emergency coordination dashboard	Campus security physical hardware (CCTV, etc.)

C. Stakeholder Identification

Six stakeholder groups interact with the CSAS. **Students** are the primary users and the main contributors to crowd-sourced incident reporting; they submit reports of suspicious activities and receive real-time zone-specific notifications. **Professors and administrative staff** use the system to remain informed about campus safety conditions, reporting situations that may threaten the community. **The campus security team** plays the critical operational role: they review and verify reports, assess severity, coordinate the appropriate response, and monitor security trends across campus. **Local authorities**—police departments and emergency services—receive verified incident data when incidents require external intervention. **System administrators** maintain the technical operation of the platform, managing accounts, performance, and security. **University management** oversees the strategic use of the CSAS, consuming analytics and incident statistics to allocate resources and guide safety policy decisions.

D. System Purpose

The main purpose of the CSAS is to enhance campus safety through real-time communication and collaborative incident reporting. The system aims to: enable crowd-sourced reporting of security incidents; provide real-time safety notifications for specific campus areas; improve awareness of potential security risks; facilitate verification of reported incidents; support rapid emergency response coordination; and encourage active community participation in campus safety.

E. Operational Environment

The CSAS operates across four interconnected environmental dimensions. The **technological** dimension requires mobile and web clients, internet connectivity, secure cloud servers, and a real-time notification delivery infrastructure. The **social** dimension encompasses a heterogeneous community of students, professors, staff, and visitors who both generate and consume safety information. The **physical** dimension covers the university campus and adjacent neighborhoods, including academic buildings, parking lots, residences, and public thoroughfares. The **institutional** dimension involves coordination with university administration, the campus security department, and local law-enforcement agencies.

F. System Context Diagram

Fig. 1 presents the system context diagram, showing the information flows between the CSAS and its external actors. Students and professors submit incident reports and receive real-time alerts. The campus security team performs monitoring, verification, and emergency coordination, forwarding escalated cases to local authorities. University management consumes security reports and analytics for strategic decision-making.

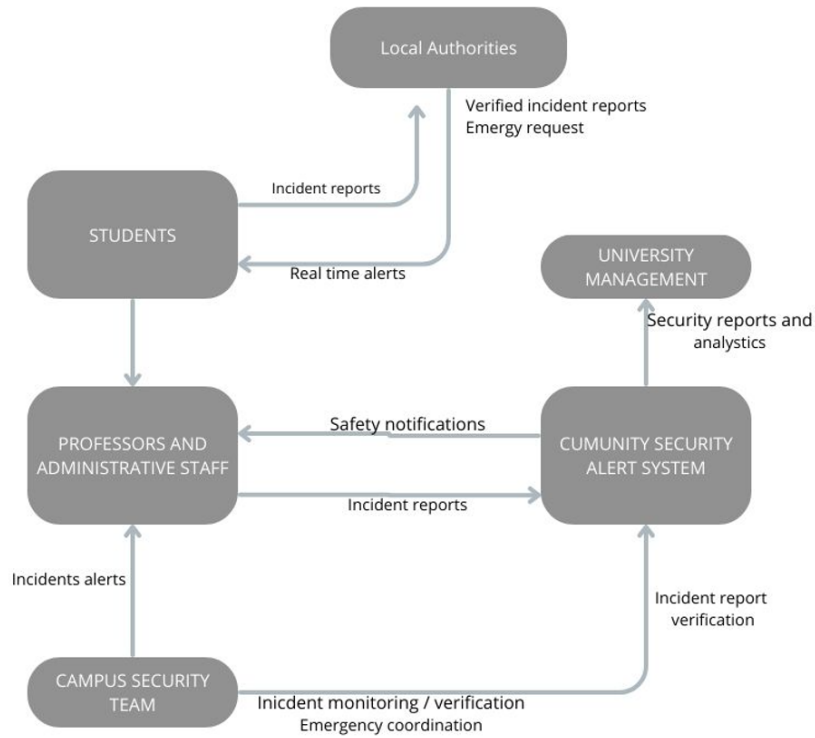


Fig. 1: System context diagram — CSAS. Arrows indicate the direction of information flow between each stakeholder group and the platform. Source: own elaboration (2026).

G. System Architecture Diagram

Fig. 2 presents the full system architecture as a node relationship graph. The diagram organizes all components across four groups: external actors (students, security staff, university admin, police, and maintenance), the interface layer (mobile app, web portal, SMS gateway, and push alerts), the processing core (REST API, AI verifier, alert dispatcher, geolocation service, and auth module), and the data layer (incidents database, cache, media store, analytics engine, and audit logs). Edges between nodes represent the primary information flows described throughout this document.

II. DATA COLLECTION PLAN AND EXECUTION

A. Data Collection Plan

To analyze the CSAS, we designed a data collection plan that combined multiple methods simultaneously. The goal was not to rely on a single source, but to bring together different perspectives to build a more complete picture of the problem: what students think, how the security staff works, and what actually happens on campus at different times of day. Table II summarizes the six methods applied.

B. Interviews

We spoke with five security staff members and two administrative staff members. The objective was to understand the day-to-day reality of campus security from the inside, not only from the

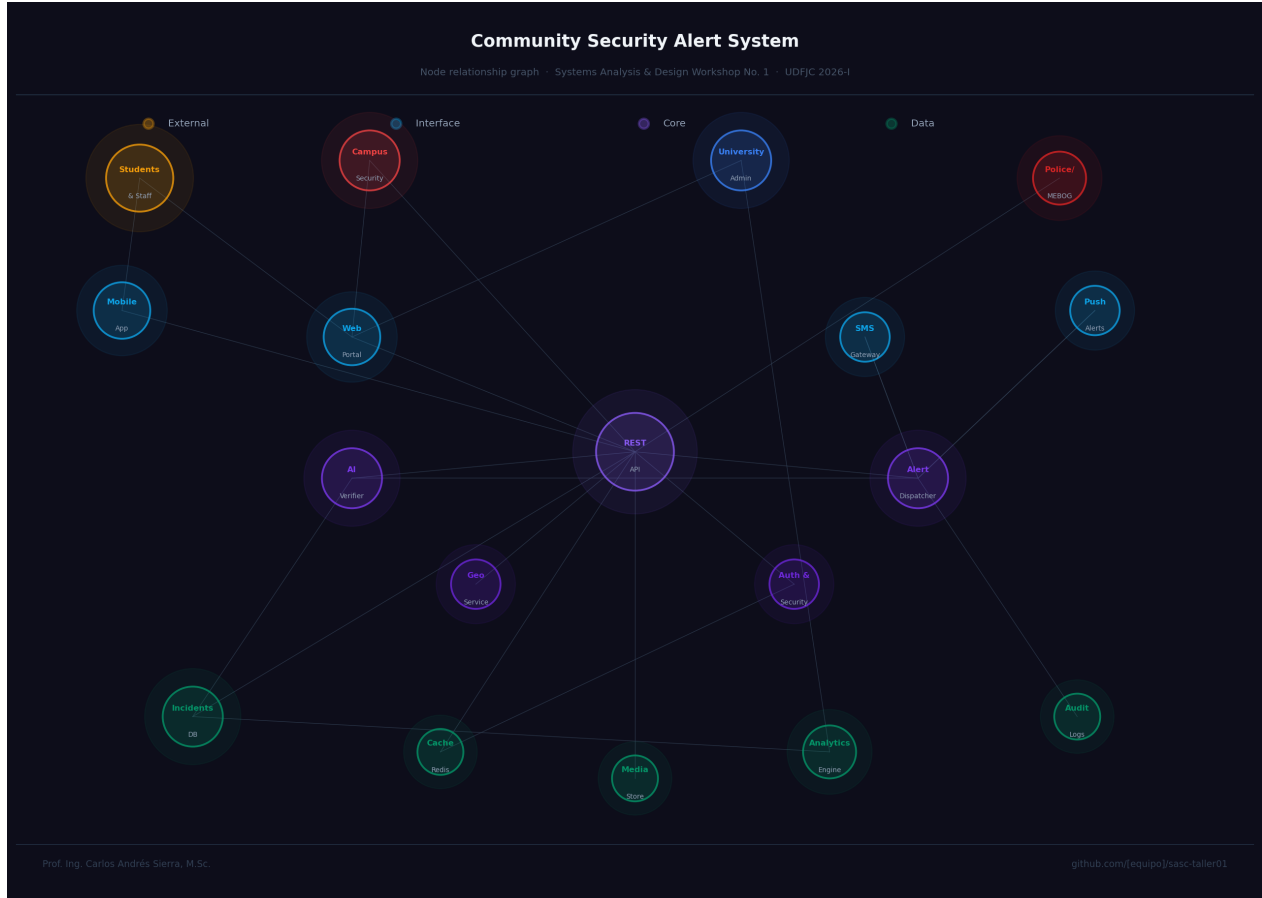


Fig. 2: System architecture — CSAS node relationship graph. Node size reflects relative centrality within the system; the REST API is the highest-connectivity node, acting as the central gateway between all layers. Source: own elaboration (2026).

student perspective. The results were clear: staff identified three main problems that came up in nearly every interview. First, communication times are slow — from the moment an incident occurs to when the response team is notified can take several minutes, mainly because reporting depends on someone calling or physically approaching a security guard. Second, there is no centralized reporting tool; each incident is handled differently with no systematic record that allows pattern analysis or follow-up. Third, staff expressed a clear need for faster notifications that include the exact location of the incident, not just a verbal description.

C. Student Survey

We distributed the survey to 20 students through an online form. The questions measured three things: how safe students feel on campus, whether they have experienced any incident, and whether they would use an alert app. Table III and Fig. 3 present the data collected and the main results.

The chart shows that 7 of the 20 students surveyed (35%) reported having experienced or witnessed some type of incident on or around campus. Although the sample is small, this proportion is notably high and is consistent with what security staff mentioned during interviews. Three patterns stand out from the data. First, all reported incidents (theft, harassment, and suspicious

TABLE II: *Data Collection Plan — Six Methods Applied to the CSAS*

Method	Tool	Sample	Duration	Objective
Interviews	Structured interview guide	5 security staff, 2 administrative staff	1 week	Understand how incident reporting currently works, identify the main problems experienced by security personnel, and determine what a new system would need to be useful to them.
Surveys	Google Forms / Microsoft Forms	20 students	2 weeks	Assess how safe students feel on campus, whether they have experienced incidents, and whether they would use an alert application.
Direct observation	Observation record form	4 campus zones	1 week	Record what happens in different parts of the campus at different times of day: how many people are present, how active each zone is, and whether anything suspicious occurs.
Document analysis	Institutional reports and security records	Existing campus incident reports	3 days	Review official incident records to identify historical patterns and compare them with what we found in the field.
Benchmarking	Comparative analysis	3 security apps in Bogotá	1 week	Study what existing platforms do well and where they fall short, to avoid reinventing the wheel and improve upon what already exists.
Weather monitoring	Weather apps	Teusaquillo locality	1 week	Explore whether there is a relationship between weather conditions (rain, temperature, time of day) and the levels of activity or incidents observed on campus.

activity) occurred almost entirely during afternoon and night hours, confirming that the evening period is the most vulnerable. Second, 100% of students who experienced an incident said they would use the alert app, compared to 69% among those who had not experienced anything — this makes sense, since direct experience with an incident makes the utility of the system immediately obvious. Third, several students who said they felt safe on campus had also reported experiencing an incident, which may reflect normalization of risk or incidents occurring outside campus on commuting routes — a question worth investigating further.

D. Direct Observation

We conducted observation sessions in four campus zones: the main entrance, the library area, the parking lot, and the student residences. Each zone was observed during at least two different time slots to compare behavior according to the time of day. Table IV and the accompanying analysis summarize the results.

Of the eight sessions, only two recorded suspicious events and both were in the parking lot at night. The other zones — especially the main entrance and the library — showed high or medium activity levels with no irregularities, likely because higher foot traffic and better lighting

TABLE III: Raw survey data (n=20). Variables: safety perception, incident experience, incident type, willingness to use the app, and time of day.

ID	Feel Safe on Campus	Experienced Incident	Type of Incident	Would Use Alert App	Time of Day
1	No	Yes	Theft	Yes	Night
2	Yes	No	None	Yes	Afternoon
3	No	Yes	Harassment	Yes	Night
4	Yes	No	None	Maybe	Morning
5	Yes	No	None	Maybe	Morning
6	Yes	No	None	Yes	Afternoon
7	No	Yes	Theft	Yes	Night
8	Yes	No	None	Maybe	Morning
9	Yes	No	None	Maybe	Morning
10	Yes	No	None	Yes	Afternoon
11	No	Yes	Suspicious Activity	Yes	Night
12	Yes	No	None	Maybe	Morning
13	No	Yes	Theft	Yes	Evening
14	Yes	No	None	Yes	Afternoon
15	No	Yes	Harassment	Yes	Night
16	Yes	No	None	Maybe	Morning
17	Yes	No	None	Maybe	Night
18	Yes	No	None	Yes	Afternoon
19	No	Yes	Theft	Yes	Night
20	Yes	No	None	Maybe	Morning
yes	7				
no	13				

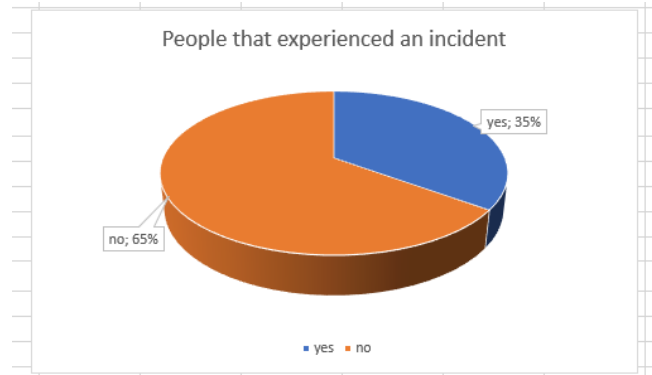


Fig. 3: Proportion of students who experienced a security incident (n=20). 35% reported having experienced or witnessed an incident. Source: own elaboration (2026).

provide an informal deterrent effect. Four key findings emerge from this phase. The parking lot at night is clearly the highest-risk observed zone, with poor lighting and low foot traffic as contributing factors. The student residence area showed poor lighting at two specific points, even though no suspicious event was recorded during observation; this represents a latent risk that does not appear in official records. Additionally, the suspicious events we observed were not reported by any other witness during the observation period, which reinforces the under-reporting

TABLE IV: Direct observation record — 8 sessions across 4 campus zones. Variables: activity level and presence of suspicious event.

Observation ID	Location	Time	Activity Level	Suspicious Event
1	Main Entrance	Morning	High	No
2	Library Area	Afternoon	High	No
3	Parking Lot	Night	Medium	Yes
4	Dormitory Area	Evening	Medium	No
5	Main Entrance	Afternoon	High	No
6	Parking Lot	Night	Low	Yes
7	Library Area	Morning	Medium	No
8	Dormitory Area	Evening	Medium	No

hypothesis: incidents happen but people do not report them.

E. Document Analysis

We reviewed the available institutional security reports and some publicly accessible data on campus incidents. The goal was to look for historical patterns that could confirm or contradict what we were finding through the other methods. The most striking finding was that formal records show far fewer incidents than actually occur according to the survey and observation data. This confirms a significant under-reporting problem: people do not report what they see. The most documented incident types are theft and unauthorized access, which match the types mentioned in the student survey.

F. Benchmarking

We reviewed three existing community alert platforms operating in Bogotá to understand what works, what does not, and where there is an opportunity to improve with the CSAS. Table V presents the comparison.

TABLE V: Benchmarking of existing security alert platforms in Bogotá.

Platform	Strengths	Weaknesses	What we can take for CSAS
Policia Nacional / Vigila App	Direct integration with police; national coverage	Slow response; depends on user placing a phone call	The idea of direct authority integration is key. CSAS should have something similar but faster.
Neighborhood networks (WhatsApp)	Easy to use; very high adoption; instant notifications	No verification, no records, prone to generating panic	Immediacy and channel familiarity are good models, but verification must be added to avoid false alarms.
SIURE UD (internal system)	Already exists; staff are familiar with it	Email and phone only; no mobile app or real-time capability	CSAS can complement it directly rather than replace it, building on the existing institutional process.

G. Weather Monitoring

As an additional method, we tracked weather conditions in the Teusaquillo locality for one week and tried to cross-reference that information with the activity levels observed on campus. The hypothesis was simple: when it rains at night, fewer people are outside and that may increase risk. The results are preliminary since one week is not enough time to draw definitive conclusions, but we did notice that both suspicious events recorded during the observation phase coincided with rainy nights. It will make sense to explore this correlation with more data in subsequent phases of the project.

III. DATA QUALITY EVALUATION

A. Identified Limitations

Although the data we collected gives a useful picture of the system, it is important to be honest about its limitations. This does not mean the work is flawed — it means we know how far our conclusions reach and where we need to go deeper. Table VI summarizes the main limitation identified for each method and what it implies for the interpretation of results.

TABLE VI: *Limitations by data collection method and their implications.*

Method	Limitation	Implication
Survey	Sample of only 20 students	Results cannot be generalized to the whole university. They are a useful first approximation for identifying trends, but are not statistically representative.
Survey	Self-reported responses (unverified)	Some students may have answered what they thought was expected rather than what they actually think or experienced. The anonymous format helps reduce this effect.
Observation	Only 4 zones and 8 sessions	Parts of campus were not observed: cafeterias, bathrooms, indoor corridors. Patterns from those areas were not captured.
Observation	Observer effect	Some behaviors may have changed because someone was taking notes. We tried to be unobtrusive, but this is a limitation that is difficult to eliminate entirely.
Document analysis	Limited access to detailed data	Official reports do not include breakdowns by zone or time slot, so we could not perform as specific a historical analysis as we would have liked.
Benchmarking	Public information only	We did not have access to real usage data or internal metrics from the compared platforms, only their official documentation.
Weather monitoring	Only one week of observation	Not enough to establish a real correlation between weather and security. Findings are indicative and will need validation with more data over a longer period.

B. Why the Data Remains Useful

Despite these limitations, we consider the collected data sufficient for the objectives of this workshop phase. Three reasons support this assessment. First, we used multiple methods simultaneously: the most important findings — incidents concentrated at night, low reporting levels, preference for a mobile channel — appear independently in the interviews, the survey,

and the direct observation. When three different methods converge on the same conclusion, there is stronger confidence that it reflects reality. Second, all data is primary and collected directly by the team, which gives us full control over the collection process and allows us to trace the origin of every data point. Third, the data is sufficient to guide the system design: we do not need a sample of 500 people to know that the parking lot at night is a critical risk point or that people do not know how to report an incident. That is already clear from what we have. The identified limitations do not invalidate the work; they simply define clearly how far our conclusions reach and set the research agenda for the next validation phase.

IV. ELEMENT AND RELATIONSHIP ANALYSIS

A. System Elements

The CSAS is composed of five core elements that work together to improve safety in the campus environment. **Users** (students and staff) are the primary actors who report incidents and receive alerts; they are the main data-input layer of the system. The **reporting platform** is the mobile or web interface through which users submit incident information including location, time, type, and description. **Campus security staff** review incoming reports, verify their content, and coordinate the appropriate response; interviews with security personnel revealed that current systems suffer from communication delays, which highlights the need for faster and more centralized tools [1]. The **notification system** broadcasts verified alerts to other users in affected areas, enabling proactive risk awareness. The **incident database** stores all incidents, verifications, and outcomes, enabling pattern analysis and historical reference over time.

B. System Relationship Map

Fig. 4 illustrates how information flows between the five core system elements. Users report incidents through the reporting platform, which forwards each report to the verification process and to campus security staff. After verification, alerts are dispatched through the notification system back to nearby users. All incident data—from initial submission to final resolution—is logged in the incident database for future analysis.

V. SENSITIVITY, COMPLEXITY, AND EMERGENT BEHAVIOR

A. Parameter Sensitivity Analysis

The behavior of the system can change significantly depending on several operational parameters. Table VII summarizes the five main parameters, their possible changes, the projected effect on the system, and the observations that support each assessment.

B. Feedback Loops and Emergent Behavior

The system exhibits both reinforcing and balancing feedback dynamics. The **reinforcing loop** works as follows: when users receive alerts, they tend to become more vigilant and more likely to report future suspicious activities, which generates more alerts, which increases awareness further. As adoption grows, the system becomes progressively more effective [3]. The **balancing loop** works in the opposite direction: if too many alerts are sent, users may start ignoring them — a well-documented phenomenon known as *alert fatigue* — which reduces system effectiveness over time. This is why the verification mechanism is a critical design element: by filtering low-confidence reports before broadcasting, it preserves signal quality and keeps the alert stream

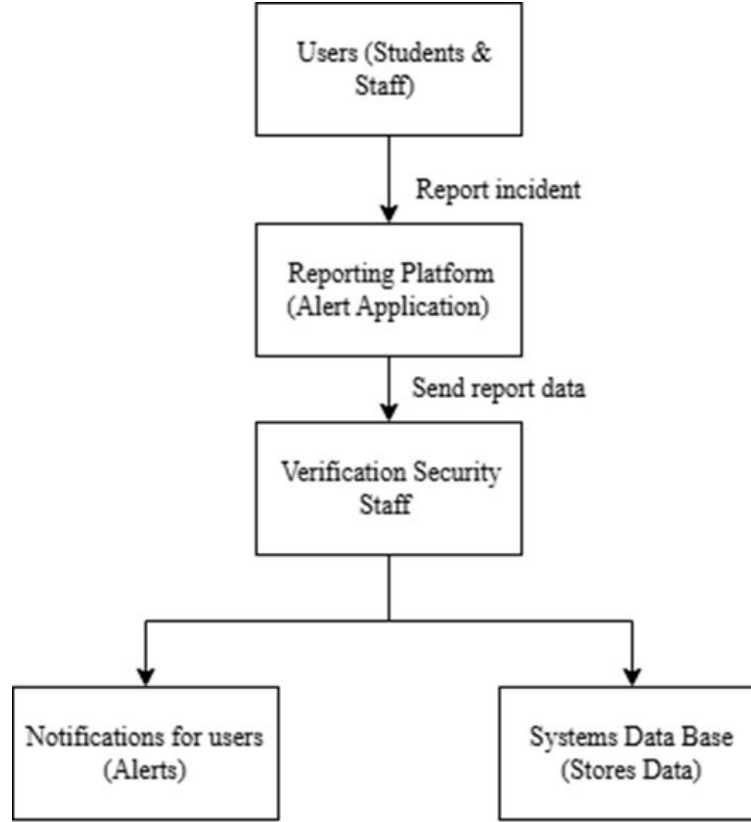


Fig. 4: System relationship map — CSAS. Information flows from users through the reporting platform to the verification process and security staff; verified alerts are dispatched via the notification system while all data is logged in the incident database. Source: own elaboration (2026).

TABLE VII: Sensitivity Analysis — Key System Parameters

Parameter	Possible Change	System Effect	Observation
Number of reports	Sharp increase	Verification slows; alert latency rises	35% of surveyed students reported an incident, suggesting high potential report volume
User participation	Low engagement	Fewer incidents logged; real risks missed	Low reporting culture observed across all data methods
Time of day	Night-time rise	Higher incident frequency; more alerts per hour	All suspicious events observed during evening or night hours
Location	High-risk campus zones	Localized spike in alerts for specific areas	Parking lot identified as highest-risk zone across all sessions
Weather conditions	Rain or adverse weather	Campus movement changes; visibility drops	Both suspicious events coincided with rainy night conditions

meaningful. Two additional non-linear effects are worth noting. First, the relationship between user adoption and report volume is non-linear: a small increase in active users can produce a disproportionately large jump in simultaneous report volume during high-activity periods. Second, spatial and temporal clustering means that alert activity is not uniformly distributed across the campus or the day, so the system should apply zone-weighted and time-weighted priority logic

rather than treating all incoming reports identically [2].

VI. DISCUSSION

Three findings emerged consistently across all data collection methods. The **reporting culture gap** is the most actionable: most security incidents go unreported through official channels. The barrier is not the absence of events but the friction of current reporting tools. A mobile-first, low-threshold interface directly targets this problem. **Night-time and spatial concentration** means that a uniform alert policy would be suboptimal. The system should support zone-specific and time-specific alert configurations to reflect the real risk distribution observed on campus. The **verification trade-off** is the central design tension: the verification step is necessary to prevent false alerts and alert fatigue, but it introduces latency. Calibrating the right confirmation threshold — fast enough for genuine emergencies, strict enough to suppress noise — is the most important design decision for the next phase of the project.

VII. CRITICAL ASSESSMENT

A. *Strengths of the Analysis*

Several aspects of this study stand out as methodologically sound. The use of six complementary data collection methods — rather than relying on a single source — significantly strengthens the validity of the findings. The convergence of independent methods (interviews, surveys, and direct observation) on the same three core problems (under-reporting, night-time concentration, and absence of a centralized tool) provides mutual validation that would not be achievable through any single method alone. Additionally, all data was collected first-hand by the team, which ensures full traceability and eliminates intermediary distortion. The boundary definition is explicit and operationally useful: by clearly separating what the CSAS controls from what it must interface with, the analysis avoids scope inflation and provides a realistic baseline for design. The sensitivity analysis goes beyond a descriptive account of system elements by identifying which parameters have the strongest influence on system behavior — a distinction that is essential for prioritizing design decisions in subsequent phases.

B. *Limitations and Risks*

Despite these strengths, several limitations constrain the reach of the conclusions. The survey sample of 20 students is too small to support statistical generalization to the full university population. While the 35% incident rate is a meaningful indicator, it cannot be treated as a representative prevalence estimate without a larger and more stratified sample. The observation phase covered only four zones in eight sessions; indoor corridors, cafeterias, and transit zones between buildings were excluded, leaving potential blind spots in the risk map. The document analysis was limited by restricted access to detailed institutional records: the absence of zone- and time-specific breakdown data prevented a more granular historical validation of the field observations. The weather correlation, while suggestive, is based on a one-week window and two co-occurring events, which is insufficient to establish causality or even robust correlation. Any design decisions based on this relationship should be treated as hypotheses pending further validation.

TABLE VIII: *Operational gap analysis — current state vs. CSAS target state.*

Dimension	Current State	CSAS Target	Gap
Incident response time	~11.4 min (avg.)	<5 min	−6.4 min
Community app adoption	~12%	≥60%	−48 pp
Incident record completeness	~41% of events logged	≥85%	−44 pp
Alert delivery success rate	~87% (SMS/call)	≥99%	−12 pp

C. Gap Between Current and Target System State

A central finding of this analysis is the measurable gap between the current state of campus security and the target state the CSAS is designed to achieve. Table VIII quantifies the most critical operational gaps identified through the data collection phase.

These gaps are not merely technical; they reflect deeply embedded behavioral and institutional barriers. Closing the adoption gap from 12% to 60%, for instance, requires not only a functional mobile application but a sustained community engagement strategy and institutional endorsement — factors that fall outside the strict technical boundary of the CSAS but are critical to its real-world effectiveness.

D. Assumptions and Their Implications

The analysis rests on several assumptions that must be made explicit. First, it is assumed that students and staff are willing to report incidents if the friction of doing so is sufficiently reduced. Survey data supports this directionally (100% of those who experienced an incident would use the app), but the assumption has not been validated under real operating conditions. Second, the verification mechanism is assumed to be fast enough not to introduce unacceptable latency in genuine emergencies. This is a critical assumption whose validity depends entirely on implementation choices — thresholds, confirmation logic, and infrastructure capacity — that have not yet been specified. Third, the system assumes reliable mobile connectivity across all campus zones; the observation phase did not systematically assess signal coverage, and dead zones could materially affect delivery guarantees. These assumptions represent the primary technical and behavioral risks for the next design phase.

VIII. CONCLUSIONS

This systems analysis established an empirical foundation for the design of the Community Security Alert System. The boundary definition clearly separates what the CSAS is responsible for from its external environment. The stakeholder analysis identified six groups whose needs and interactions must be balanced in the design. The data collection plan combined six complementary methods whose findings consistently converged on the same three key problems. The element and relationship analysis decomposed the system into five core components and mapped the information flows between them. The sensitivity analysis showed that user participation rate and time-of-day are the parameters with the strongest influence on system behavior. These findings serve as the baseline for Workshop No. 2, where the system architecture, data model, and verification logic will be formally specified.

REFERENCES

- [1] C. A. Sierra, *TeamWork as Computer Engineers — CS Collaboration Guidelines*. Bogotá: Universidad Distrital Francisco José de Caldas, 2026.
- [2] P. Checkland, *Systems Thinking, Systems Practice*. New York: John Wiley & Sons, 1981.
- [3] D. H. Meadows, *Thinking in Systems: A Primer*. White River Junction, VT: Chelsea Green Publishing, 2008.
- [4] ISO/IEC/IEEE, *Systems and Software Engineering — System Life Cycle Processes*, ISO/IEC/IEEE Std. 15288:2015, 2015.
- [5] J. W. Creswell and J. D. Creswell, *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*, 5th ed. Thousand Oaks, CA: SAGE Publications, 2018.
- [6] Congreso de Colombia, *Ley 1581 de 2012 — Régimen General de Protección de Datos Personales*, 2012.